ENGINEERS WALLAH

CHEMISTRY

Lecture VIII

Atomic Models and 'Iso' Species

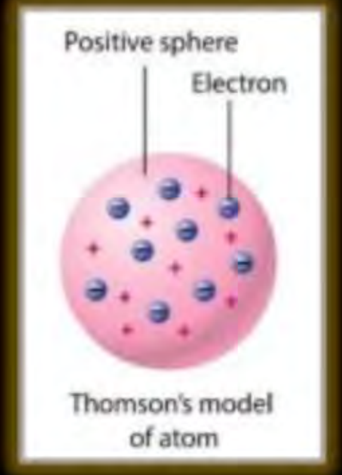


By: Sunil Bhaiya

TOPICS *to be covered*

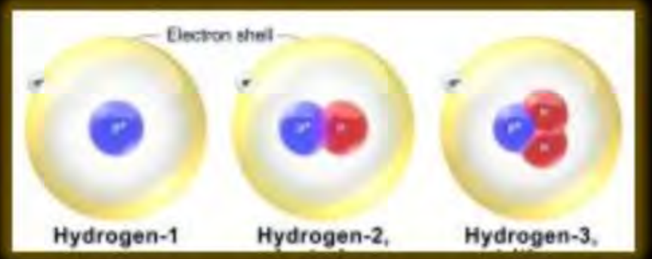
1. Atomic Models in One Shot
2. Master All 'Iso' Species in One Shot

Knowledge Ride On



Atomic Models in One Shot

Knowledge Ride On



Master All 'Iso' Species in One Shot

Knowledge Ride On



Efficiency Hacks ✓



Glossary

Atomic number (परमाणु संख्या) ✓
Mass number (द्रव्यमान संख्या) ✓
Isotopes (समस्थानिक) ✓
Isobars (समभारिक) ✓
Isotones (समन्यूट्रॉनिक) ✓
Isoelectronic species (समइलेक्ट्रॉनिक) ✓

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following is not a colloid?

RRB JE 2015

		D.P.	D.M.	Type of colloid
A	Chlorophyll (complex of magnesium ion)			
B	Smoke	Solid	Gas	Solid aerosol
C	Ruby glass	Solid	Solid	Solid sol
D	Milk	Liquid	Liquid	Emulsion

Question

The number of molecules present in 36 grams of water is:

RRB JE 2015

A 3.023×10^{23}

B 6.046×10^{23}

C 3.023×10^{46}

D 12.046×10^{23}

Remember the Below Line

1 molecule of H_2O
↓
$= 2 \times 1 + 16$
$= 18u$

1 mole atoms/molecules/ formula units = **6.02×10^{23} atoms/molecules/formula units** = **Gram atomic/Gram molecular/Gram formula unit mass**

6.02×10^{23} molecules = 18 g

x molecules = 36 g

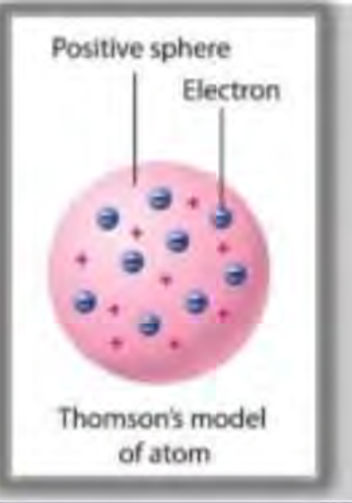
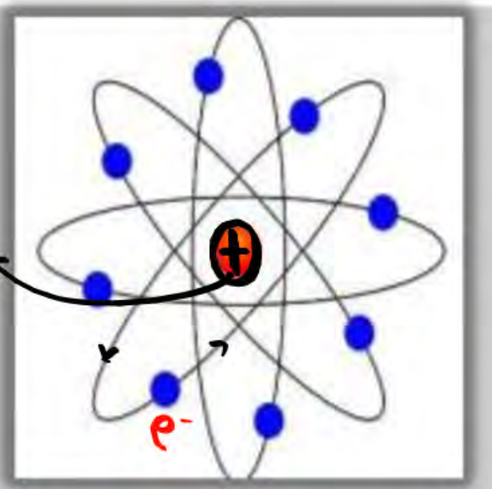
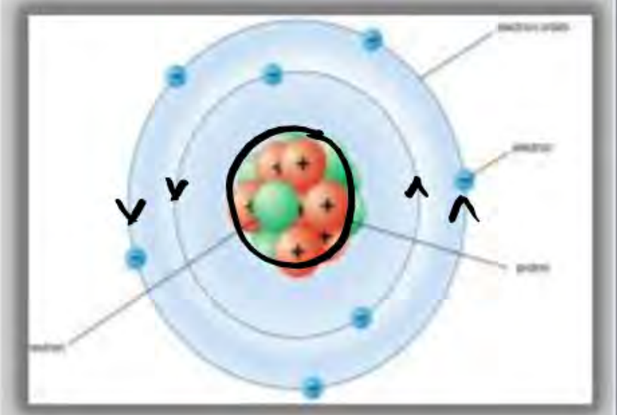
$$\Rightarrow \frac{6.02 \times 10^{23} \times 36\ g}{18\ g} = x$$

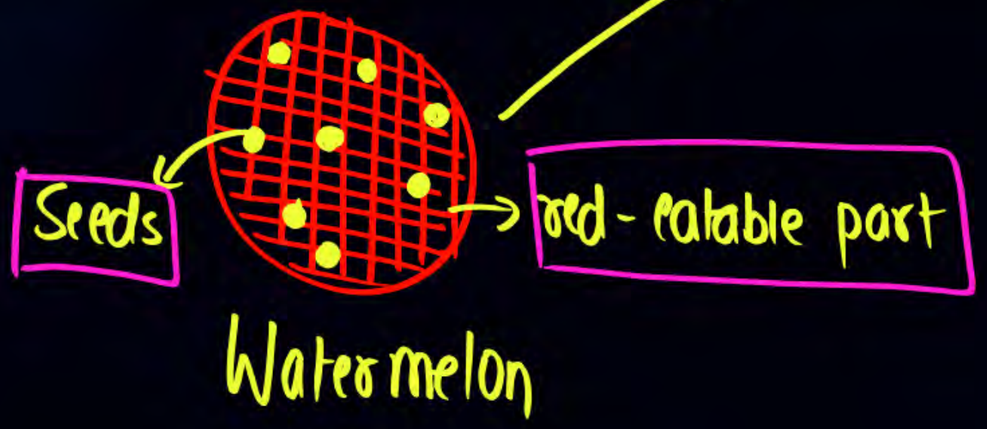
$$\Rightarrow x = 12.04 \times 10^{23}$$

Concept Bites – Atomic Models in One Shot

Property/Feature	Thomson's Atomic Model (1904)	Rutherford's Atomic Model (1911)	Bohr's Atomic Model (1913)
Other name/Special feature	Plum-pudding model or watermelon model	Nuclear model or planetary model of an atom	First atomic model that explains stability of an atom.
Important Points	- An atom consists of positively charged sphere and electrons are embedded in it. - The mass of an atom is distributed uniformly throughout it.	- An atom consists of positively charged nucleus at the centre where most of the mass of an atom is concentrated.) - The electrons revolve around the nucleus in circular orbits.	- An atom consists of small positively charged nucleus in the centre and the electrons revolve around it in circular paths called discrete orbits (separate)

Discrete orbit
↳ (fixed energy & fixed radius)

Property/Feature	Thomson's Atomic Model	Rutherford's Atomic Model	Bohr's Atomic Model
Representation of Atomic Model			





Rutherford's Alpha Scattering Experiment

1908-1913

Hans Geiger & Ernest Marsden
↳ under guidance of Rutherford

Alpha particle → CONCEPT IInd

Concept - II

$$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + \alpha\text{-particle}$$

α-particle → 2 protons & 2 neutrons

CONFIGURATION IS SAME

Helium (He)

A. No. → 2

no. of protons → 2

–"– electrons → 2

–"– neutrons → 2

→ eject 2 electrons from it (practically not feasible)

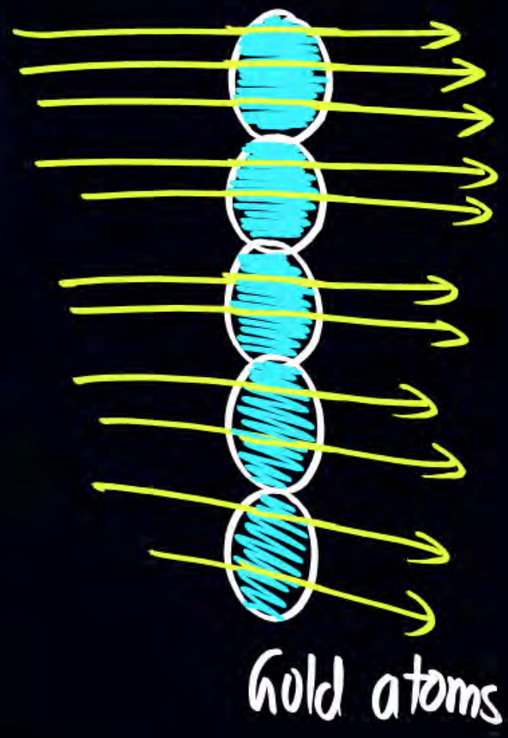
→ no. of protons → 2
–"– neutrons → 2
–"– electrons → 0

→ He^{2+} ions

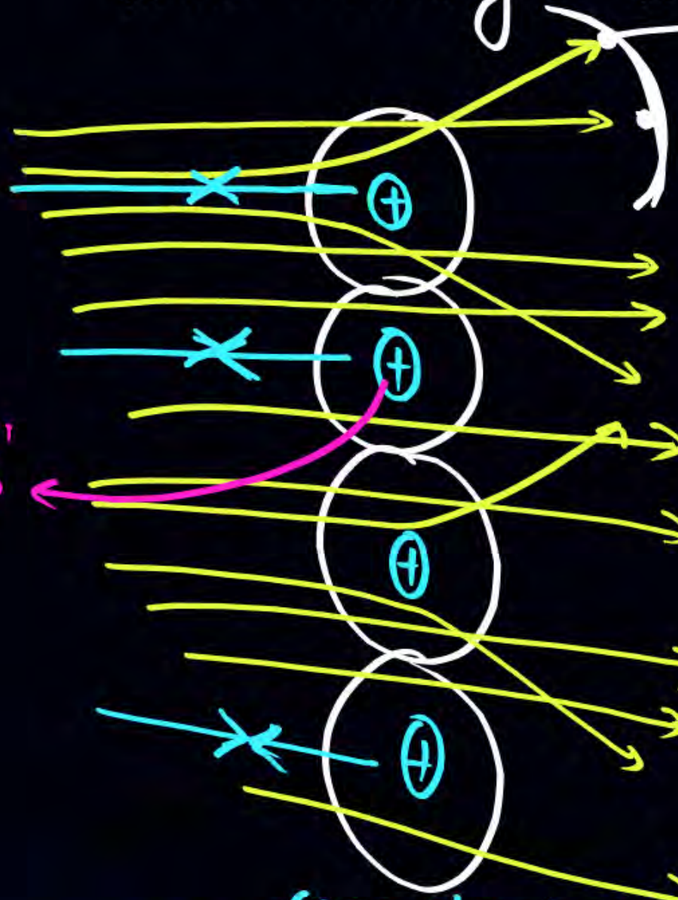
Rutherford's Alpha Scattering Experiment

Gold foil (most malleable metal & its thickness won't impact path of α particle)

If Thomson's atomic model is correct



What actually happened?



The thing that is present at centre of an atom:

- ⊕ tvely charged
- ⊕ Most of mass of an atom is concentrated at centre.

Practice and PYQ's Wallah



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Question

Rutherford's alpha scattering experiment resulted in the discovery of _ _ _ _ _ _ _.

A Electron

B Proton

C Neutron

D Nucleus

Question

Which among the following is correct for an alpha particle?

A Electrons: 2, Neutrons: 2

B Neutrons: 2, Protons: 2

C Electrons: 2, Protons: 2

D Proton: 1, Electron: 1

Question

An alpha particle is identical to:

A H^+ ions

B He^{2+} ions

C He^+ ions

D H^{2+} ions

Concept Bites – Master All 'Iso' Species in One Shot

Isotopes (समस्थानिक)

↳ same / equal

They are the atoms of the same elements which have same atomic number (परमाणु संख्या) but different mass number (द्रव्यमान संख्या).



$A = Z + n$



$A' = Z + n'$

Isotopes have:

Same no. of protons

— " — electrons

different no. of neutrons

Isotopes (समस्थानिक) of Hydrogen

Isotope	Atomic or proton number	Number of neutrons
Protium (P)	1	0

or Hydrogen (H)

no. of protons & electrons → 1

$A = Z + n$
$= 1 + 0 = (1)$

Isotope	Atomic or proton number	Number of neutrons
Deuterium (D)	1	1

no. of protons & electrons → 1

$A' = Z + n'$
$= 1 + 1 = (2)$

Isotope	Atomic or proton number	Number of neutrons
Tritium (T)	1	2

$A'' = Z + n'' = 1 + 2 = (3)$

$_1^1H$ or $_1H^1$

$_1^2H$ or $_1H^2$

$_1^3H$ or $_1H^3$

Isotopes (समस्थानिक) of Chlorine

$^{35}_{17}Cl$ & $^{37}_{17}Cl$

Isotope	Atomic or proton number	Number of neutrons
Cl-35	17	18

no. of protons & electrons → 17

$A = Z + n$

$= 17 + 18 = 35$

Isotope	Atomic or proton number	Number of neutrons
Cl-37	17	20

$A' = Z + n'$

$= 17 + 20$

$A' = 37$

Isobars (समभारिक)

↳ Same/equal

They are the atoms of different elements which have different atomic number but same mass number.

$^{40}_{18}Ar$ & $^{40}_{20}Ca$

Isobar	Atomic or proton number (Z)	Number of neutrons
Argon (Ar)	18	22

no. of protons → 18 & electrons

$A = Z + n$
$= 18 + 22 = 40$

no. of proton & electron = 20

Isobar	Atomic or proton number (Z)	Number of neutrons
Calcium (Ca)	20	20

$A = Z' + n' = 20 + 20$
$= 40$

Isotones (समन्यूट्रॉनिक)

↳ Same/equal

They are the atoms of different elements which have different atomic and mass number but have the same number of neutrons.

Isotopes of C: $^{12}_{6}C$, $^{13}_{6}C$ & $^{14}_{6}C$

$^{14}_{6}C$

Isotone	Atomic or proton number (Z)	Mass number (A)
Carbon (C)	6	14

no. of proton & electron → 6

$A = Z + n$

$14 = 6 + n$

$n = 8$

– II → 8

Isotone	Atomic or proton number	Mass number
Oxygen (O)	8	16

$A' = Z' + n$

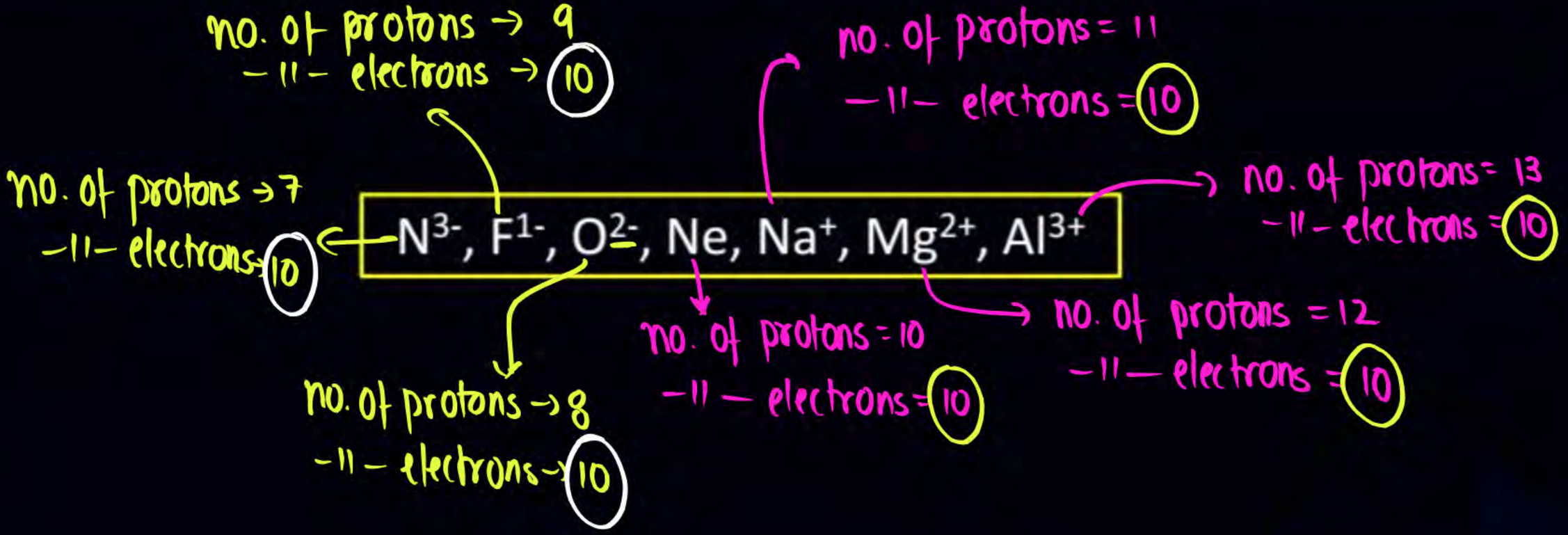
$16 = 8 + n$

$\Leftarrow$ $n = 8$

Isoelectronic Species (समइलेक्ट्रॉनिक)

Same/equal

These chemical species contain the same number of electrons.



Practice and PYQ's Wallah

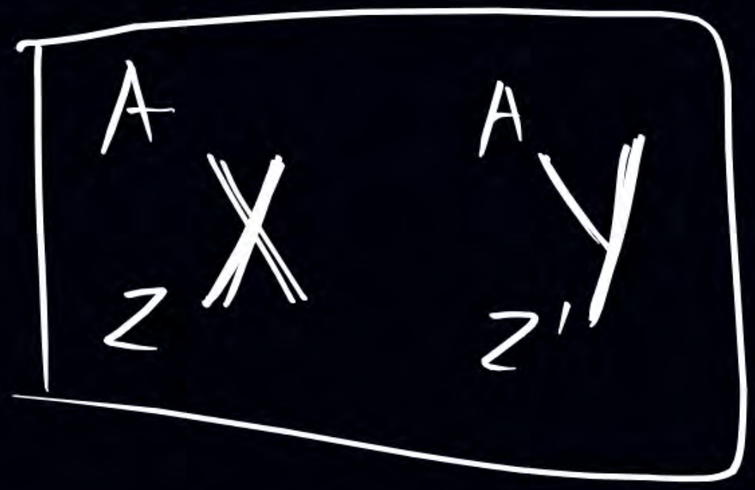


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Question

The elements which have the same mass number but different atomic numbers are known as:

RRB JE 2014



A Isotones

B Isobars

C Isotopes

D Halogens

Question

Isotopes of same element have_____.

RRB JE 2014

A Same number of neutrons ✗

B Same atomic mass ✗

C Same number of protons ✓

D Different atomic number ✗

Question

Isotopes of hydrogen are:

RRB JE 2019

A $_{2}H^{1}, _{2}H^{2}, _{1}H^{3}$

B $_{1}H^{1}, _{1}H^{2}, _{1}H^{3}$

C $_{1}H^{1}, _{1}H^{2}, _{2}H^{1}$

D $_{1}H^{1}, _{1}H^{2}, _{3}H^{1}$

Question

RRB SSE 2015

Neon has three natural isotopes with mass number 20, 21 and 22. The relative abundance of the isotopes 90.51%, .27% and 9.22% respectively. The average atomic mass of neon will be:

A 22.0 u

B 21.179 u

C 22.800 u

D 20.187 u

Weighted average

$$\Rightarrow \frac{20 \times 90.51}{100} + \frac{21 \times .27}{100} + \frac{9.22 \times 22}{100} = 2.2$$

$$\Rightarrow 18.10 + .056 + 2.2$$

$$\Rightarrow 20.3\ldots$$

Efficiency Hacks!

Ways to Improve Concentration

- Reduce multi-tasking
- Freeze slot for everything that you do in your day
- Sleep well
- Eat well




#sbsathhai
#pwsathhai
Thank
You !